

Ai-Da's inaugural exhibition "Unsecured Futures" will run from June 12 through July 6, 2019, at The Barn Gallery, St John's College, University of Oxford. The exhibition will showcase Ai-Da's drawings and performance art, including a work that references, and is an homage, to Yoko Ono's Cut Piece.

Alongside this, emerging artists work together to create works in plastic, silver and bronze 2D and 3D, developed through AI processes and new technologies. Ai-Da and this exhibition are set to dazzle and entertain, but more importantly, get us to think and engage, and help us consider our world a little more closely.

A 3D printer powered by machine vision and artificial intelligence

Objects made with 3D printing can be lighter, stronger and more complex than those produced through traditional manufacturing methods. But several technical challenges must be overcome before 3D printing transforms the production of most devices.

Commercially available printers usually offer only high speed, high precision or high-quality materials. Rarely do these offer all three, limiting their usefulness as a manufacturing



The first printer made by Inkbit uses machine vision and artificial intelligence to expand the range of materials it can print with (Credit: Inkbit)

tool. Today, 3D printing is used mainly for prototyping and low-volume production of specialised parts.

Now Inkbit, a startup out of Massachusetts Institute of Technology (MIT), is working to bring all the benefits of 3D printing to a range of products that have never been printed before—and it is aiming to do so at volumes that would radically disrupt production processes in a variety of industries.

The company is accomplishing this by pairing its multi-material inkjet 3D printer with machine vision and machine learning systems. The vision system comprehensively scans each layer of the object as it is being printed to correct errors in real time. The machine learning system uses that information to predict the warping behaviour of materials and make more accurate final products.

According to Davide Marini, Inkbit's co-founder and chief executive officer, the company was born out of the idea of endowing a 3D printer with eyes and brains. This idea unlocks a range of applications for Inkbit's machine, which can print more flexible materials much more accurately than other printers. If an object, including a computer chip or some other electronic component, is placed on the print area, the machine can precisely print materials around it. And when an object is complete, the machine keeps a digital replica that can be used for quality assurance.

Artificial intelligence tool that can tell babies' cries apart

Researchers from Northern Illinois University have developed an artificial intelligence (AI)-based tool that can identify and distinguish between normal cry signals of babies and abnormal ones, such as those resulting from an underlying illness. The method promises to be useful to parents at home as well as in healthcare settings, as doctors may use it to discern cries among sick children.

According to the scientists, while each baby's cries are unique, they all share some common features when the cries result from the same reasons. Identifying hidden patterns in the cry signal has been a major challenge, and AI applications have now been shown to be an appropriate solution within this context.

The new research uses a specific algorithm based on automatic speech recognition to detect and recognise the features of infant cries. To analyse and classify those signals, the researchers used compressed sensing as a way to process Big Data more efficiently.

Compressed sensing is a process that reconstructs a signal based on sparse data. It is especially useful when sounds are

recorded in noisy environments, which is where baby cries typically take place.

The team has designed a new cry-language-recognition algorithm that can distinguish the meanings of both normal and abnormal cry signals in a noisy environment. The algorithm is independent of the individual crier, meaning that it can be used in a broader sense in practical scenarios as a way to recognise and classify various cry features and better understand why babies are crying and how urgent the cries are.

Lichuan Liu, associate professor at Northern Illinois University, says, "Like a special language, there are many health-related information in various cry sounds. Differences between sound signals actually carry the information. These differences are represented by different features of the cry signals. To recognise and leverage the information, we have to extract the features and then obtain information from it."

The researchers hope that findings of their study could apply to several medical care circumstances in which decision-making relies heavily on experience.